

Supplementary Material

Physics Is Already There: Rethinking Physical Injection in Latent World Models

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A Training and Evaluation Details

Reproducibility. All training configurations, per-run logs, evaluation protocols, and the exact numbers behind every table and figure are archived alongside the code; the study builds on public assets (LeWM, PhyWorld, Physion, Physion++). We will release the evaluation suite, including the per-horizon/per-partition audit tools and the random-encoder controls.

Hyperparameters. All PhyWorld runs: AdamW, lr 5×10^{-5} (from-scratch reruns: 2×10^{-5}), weight decay 10^{-3} , bf16, gradient clip 1.0, 20 epochs (60 for from-scratch cells; the pretraining-regime 2×2 uses 60 epochs uniformly), batch size 64 for free rollout ($H=8$), 128 for teacher forcing ($H=1$), 48 for $H \geq 16$. History size 3; embedding dimension 192; SIGReg weight 0.09. Seeds: 3072/1234/42 where seed-replicated. Physion++ runs use the readout split (800 clips) with the same optimizer.

OOD partitions. ID: radius/mass $\in [0.7, 1.5]$, velocity $\in [1.0, 4.0]$. OOD draws radius/mass from $[0.3, 0.6] \cup [1.5, 2.0]$ and velocity from $[0, 0.8] \cup [4.5, 6.0]$. Collision uses the four-column variant (two objects’ masses and velocities). Training sets are ID-only (1,000 trajectories); evaluation covers all four partitions ($\geq 1,600$ trajectories per domain).

Probe protocol. Ridge regression ($\alpha=1$), trajectory-level 80/20 splits, probing the encoder output (not the projector), pretrained “paper” initialization, and $K=4$ stacked consecutive latents for velocity. Each choice matters: the naive combination (random init, single frame, through-projector) reads $\rho=0.166$ for uniform velocity where the corrected protocol reads 0.939 (Trap 4, Appendix E). Random-encoder and pixel-statistics controls are run alongside every probe.

B Full Injection Results

Table 1 is the numeric form of Figure 2 of the main paper, with the seed annotations spelled out.

Probe/slot factorial (uniform, free rollout). Latent both-ODD nMSE / pixel both-ODD PSNR (dB): baseline 0.131 / 20.41; probe-only 0.167 / 19.83; slot+reweight 0.114 / 21.30; probe+slot+reweight 0.125 / 20.02. The combination underperforms the slot alone on both scales; the latent and pixel verdicts agree.

	uniform (both-ODD)	parabola (r/m-ODD)	collision (both-ODD)
Free-rollout baseline	0.131	0.127	0.393
Fixed slot	0.183	0.156	0.651
+reweight ($w=30$)	0.114 ^s	0.160	0.596
+velocity	0.207	0.096^m	0.621
Probe	0.167	0.115 ⁿ	0.647
+slot+reweight	0.125 ^s	0.127	0.607
Kinematic (free MLP)	0.155	0.178	0.560
Kinematic (strict $a=g$)	0.206	0.173	0.559
Consistency	0.151	0.147	0.57–0.64
Label-free prior	0.171	0.172	0.653
Grounded prior	0.166	0.156	0.525

Table 1: The 30-cell scan (hardest clean split, latent nMSE ↓). ^mThe one seed-robust gain: 0.096 ± 0.007 over three seeds vs. baseline 0.122 ± 0.007 , non-overlapping. ⁿRefuted by seeds: three-seed 0.137 ± 0.034 vs. baseline 0.122 ± 0.007 —the single-seed low was seed luck. ^sAlso refuted by seeds: reweighted slot 0.132 ± 0.017 and probe+slot 0.142 ± 0.017 vs. baseline 0.136 ± 0.008 (parity). Remaining cells are single-seed but sit 5–20 seed-std above baseline.

Slot reweighting sweep (uniform). Slot weight 1/30/100: 0.183 / 0.114 / 0.136 (single seed), with the $w=30$ cell seed-replicated at 0.132 ± 0.017 against the baseline 0.136 ± 0.008 . Kinematic head on the reweighted slot: velocity-ODD decoded position $\rho=0.991/0.987$ (x/y); uniform pixel horizon-28 23.34 vs. 22.09 dB (+1.25); appearance-ODD remains the weak axis (decoded ρ $0.29 \rightarrow 0.43$, vs. 0.96 without the kinematic head).

Bypass probe details. Protocol for §4.3 of the main paper, which reports the numbers. Two models are probed: the free-rollout baseline (no injection) and the slot-pinned model (structpos, $w=30$). For each, a ridge probe is fit on the dimension subset alone—the full 192, the black box [2:192], the slot [0:2], or a random pair as control—with train/test trajectories split 80/20, so decodability is measured out of sample. Values are Pearson ρ against ground-truth position on the both-ODD partition, averaged over both coordinates, from frame embeddings rather than rolled-out latents. The slot absorbs position (its supervision loss falls $2.53 \rightarrow 0.015$).

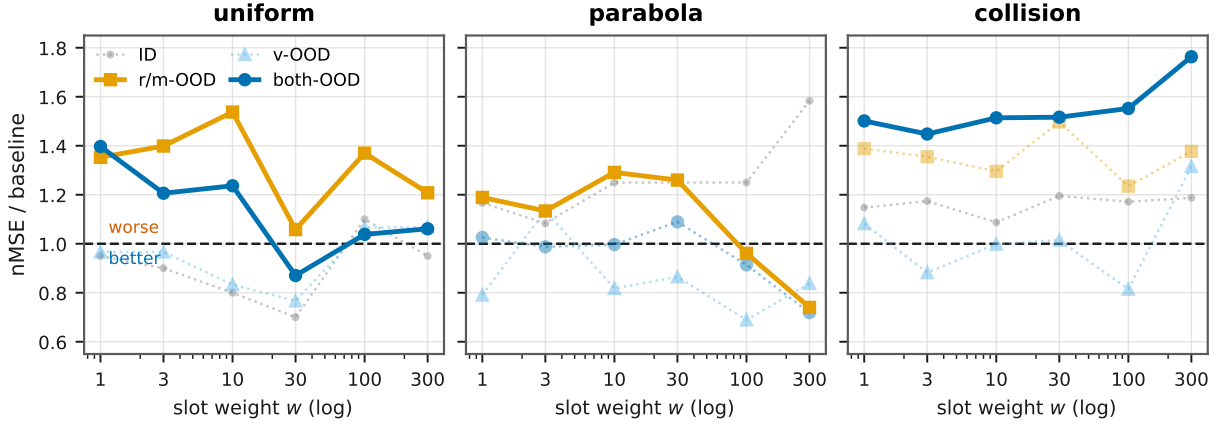


Figure 1: Dose-response re-weighting, all partitions. Solid curves are the four decision cells (hardest clean split); dotted are partitions we do not judge on—ID is not an extrapolation test, and parabola’s v-OOD/both-OOD nMSE is denominator-degenerate (Appendix E). Only 2 of the 4 decision cells reach parity (uniform both-OOD, parabola r/m-OOD); uniform r/m-OOD and collision both-OOD stay above baseline at every weight, and collision degrades as weight grows. The sweep is single-seed; both parity cells were re-run over three seeds and land at baseline (§4.3 of the main paper).

without displacing the black box’s redundant copy.

Encoded quantity vs. domain (position monotone, velocity sign-flipping). The reweighted slot’s effect is systematic in what it encodes (hardest clean split; baseline: uniform both-OOD 0.131, parabola r/m-OOD 0.127, collision both-OOD 0.393):

slot content	uniform	parabola	collision
position (structpos)	0.132	0.160	0.596
position+velocity	0.207	0.096	0.621
Δ (adding velocity)	+0.075	−0.064	+0.025

Position is an output variable with the same role in every domain, so pinning it degrades monotonically with dynamics complexity and never flips sign. The marginal effect of adding velocity (+0.075/−0.064/+0.025) tracks velocity’s role in each domain—redundant constant (uniform), linear driver (parabola), discontinuous jump (collision). The parabola gain ($0.122 \pm 0.007 \rightarrow 0.096 \pm 0.007$, every seed lower) is thus the sole seed-robust structural gain in the grid: a fixed form landing on a matching domain, not a portable prior.

Physion++ per-scenario (all ten scenarios, FR at 20 epochs). Rollout nMSE: bouncy_platform 0.041, bouncy_wall 0.094, friction_cloth 0.071, friction_collision 0.062, friction_platform 0.041, mass_collision 0.083, mass_dominoes 0.058, mass_waterpush 0.122; deformable scenes are the common hard case for every method (deform_clothhang 1.375, deform_clothhit 0.772). Injection variants degrade the rigid-body scenarios by 3–10 $\times$ (Table 2) while leaving the deformable failure in place.

Physion++ horizons. Overall nMSE at horizons 16/32/64: baseline ($H=8$) 0.016/0.140/0.280; $H=20$ 0.022/0.033/0.220; $H=16$ +scale 0.010/0.035/0.136; $H=20$ +scale 0.012/0.024/0.087; $H=28$ +scale reaches

Scenario	FR	+slot	+cons.	+cons.+acc.
friction_platform	0.041	0.119	0.078	0.067
mass_collision	0.083	0.510	0.682	0.696
mass_dominoes	0.058	0.452	0.586	0.597
mass_waterpush	0.122	0.363	0.663	1.476

Table 2: Physion++ direct training: per-scenario rollout nMSE ($\downarrow$) at 20 epochs. Injection degrades realistic-simulation prediction by 3–10 $\times$ on the mass/friction scenarios shown (deformable-object scenes are the hardest category for all methods, including the baseline). Single-seed; the gaps far exceed baseline seed noise.

horizon-64 nMSE 0.014. Free rollout is the main long-horizon lever (horizon-32 cosine 0.91 vs. 0.79–0.87 for every physics-augmented variant); geometric scale jitter acts as an amplitude stabilizer for long rollouts: without it, $H=20$ explodes on friction-collision (nMSE 24.6 at cosine 0.99); with it, 0.019.

C Cross-Backbone Replication: Frozen DINOv2

The cross-backbone variant (§3.5 of the main paper) freezes a DINOv2-small encoder (Oquab et al. 2024)—in the style of DINO-WM (Zhou et al. 2025)—and trains only the projector (as adapter) and the identical predictor stack, losses, and seeds (~ 130 runs). Free-rollout baselines (three-seed): uniform both-OOD 0.427 ± 0.039 , parabola r/m-OOD 0.225 ± 0.019 , collision both-OOD 0.479 ± 0.009 ; Physion++ horizon-64 0.258 ± 0.005 vs. teacher forcing 1.030 ± 0.028 .

Teacher forcing vs. free rollout, per domain. Values behind Figure 4 of the main paper (hardest clean

split, latent nMSE $\downarrow$, three-seed mean $\pm$ sample std). LeWM: uniform $0.300\pm0.007 \rightarrow 0.136\pm0.008$ ($2.2\times$), parabola $0.443\pm0.048 \rightarrow 0.122\pm0.007$ ($3.6\times$), collision $1.152\pm0.048 \rightarrow 0.479\pm0.079$ ($2.4\times$), Physion++ horizon-64 $1.174\pm0.041 \rightarrow 0.141\pm0.018$ ($8.3\times$). Every teacher-forcing interval is disjoint from its free-rollout counterpart. The frozen-DiNOv2 counterparts above give 1.4–4.0 $\times$ over the same four settings, so the lever survives replacing a trainable ViT-tiny with a frozen general-purpose encoder.

Injection scan on the frozen-DiNOv2 backbone
(nMSE / free-rollout baseline; vermilion = worse)

[slot] structpos	$1.17\times^\dagger$ (0.502)	$1.02\times^\dagger$ (0.229)	$0.97\times^\dagger$ (0.463)
[slot] +reweight	$0.81\times^\dagger$ (0.347)	$1.05\times^\dagger$ (0.236)	$1.01\times^\dagger$ (0.485)
[slot] +velocity	$0.93\times^\dagger$ (0.398)	$1.01\times^\dagger$ (0.227)	$0.99\times^\dagger$ (0.473)
[probe] probe	$1.02\times^\dagger$ (0.434)	$1.03\times^\dagger$ (0.231)	$1.03\times^\dagger$ (0.493)
[probe] +slot	$0.91\times^\dagger$ (0.390)	$1.10\times^\dagger$ (0.247)	$1.02\times^\dagger$ (0.487)
[dyn] free MLP	$0.99\times^\dagger$ (0.423)	$1.04\times^\dagger$ (0.234)	$1.11\times^\dagger$ (0.530)
[dyn] strict a=g	$1.08\times^\dagger$ (0.459)	$1.16\times^\dagger$ (0.261)	$1.27\times^\dagger$ (0.610)
[cons] consistency	$1.00\times^\dagger$ (0.428)	$1.01\times^\dagger$ (0.227)	$0.96\times^\dagger$ (0.461)
[free] label-free	$0.96\times^\dagger$ (0.411)	$1.00\times^\dagger$ (0.224)	$1.07\times^\dagger$ (0.512)
[free] grounded	$1.11\times$ (0.473)	$1.13\times$ (0.254)	$1.27\times$ (0.607)
	uniform (both-OOD)	parabola (r/m-OOD)	collision (both-OOD)

Figure 2: The injection scan repeated on the frozen-DiNOv2 backbone (nMSE over the free-rollout baseline; mean over available seeds, $\dagger$ = three seeds). 10 cells worse / 17 parity / 3 better (27/30 at or below baseline). The dynamics-head variants are the worst here too (strict $a=g$ and the grounded prior reach $1.27\times$ on collision); the three above-parity cells, all on uniform, are the re-weighting confound of Trap 5.

Presence and bypass. The frozen encoder, which has never seen PhyWorld and received no physics supervision or gradient, decodes position from real-frame embeddings at $\rho=0.951$ (both-OOD)—*higher* than the PhyWorld-trained LeWM encoder (0.899). The 190 non-slot dimensions alone decode at 0.951, indistinguishable from the full latent; pinning the slot at weight 300 leaves the black-box copy at 0.973. Presence and the bypass are properties of generic visual representations.

Inert, not harmful. Unlike the trainable-encoder regime, the core variants (re-weighted slot, probe, consistency) land within the baseline’s seed band on all three domains, while the plain slot ($1.42\times$ on uniform) and the dynamics-head variants (1.15 – $1.29\times$ on all domains) still hurt. With the encoder frozen, injection losses can only reshape the $384\rightarrow192$ adapter: the bypass cannot be displaced, which bounds the

Physion++ (photorealistic simulation): longer training rollout $\rightarrow$ better long horizon

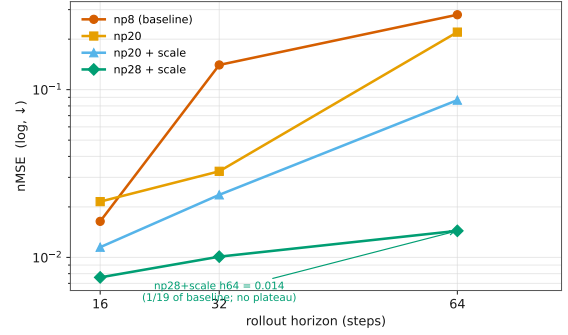


Figure 3: Physion++ direct training: per-horizon rollout nMSE (log scale) as the training horizon grows ($H=8\rightarrow28$, +scale jitter). Long-horizon error falls monotonically to 1/19 of baseline with no inflection: realistic-simulation dynamics reward long-rollout training more than the synthetic domains do.

damage—and forecloses the benefit. Injection-variant variance also roughly doubles (±0.087 vs. baseline ±0.039 on uniform), so single-seed cells on this backbone are uninformative.

D Augmentation: a Domain-Specific Lever with a Reversal Boundary

Augmentation is not part of the paper’s main line (it is neither an injection mechanism nor a protocol variable), but its boundary behavior is a useful caution. On synthetic domains, per-trial appearance jitter (brightness/contrast, strength 0.5) roughly halves the hardest split: uniform $0.131\rightarrow0.068$ (-48%), parabola -63% (r/m-OOD $0.127\rightarrow0.065$); geometric scale jitter, only in combination with a long horizon, sets the collision record ($0.393\rightarrow0.172\pm0.004$ over three seeds). Interactions are non-monotone: appearance conflicts with long-horizon training (0.472, worse than either alone), scale synergizes (0.208), the triple is worse than the best pair (0.253); temporal-stride augmentation loses even to appearance on velocity-OOD (0.556 vs. 0.376), leaving unseen velocities the open hard case.

The appearance recipe reverses on realistic simulation: the same jitter that halves synthetic OOD error degrades Physion++ friction-collision prediction by two orders of magnitude (nMSE $0.062\rightarrow6.44$) while the *aggregate* long-horizon cosine improves (Trap 1 below), and it does not help zero-shot transfer either (0.597, below the 0.607 random-prior ceiling). The boundary is principled: appearance jitter encodes an appearance-invariance assumption that holds under PhyWorld’s simple renderer but breaks where appearance carries physical information (friction, mass, material). Scale jitter, which assumes only size-invariance, keeps helping at realistic-simulation scale (Appendix B); appearance jitter flips sign. Augmentation gains are domain- and type-specific, not portable.

E Metric Pathologies and Evaluation Traps

Our anatomy repeatedly reached conclusions opposite to those suggested by common metrics; the decision rules of §4.1 of the main paper distill the following five failure modes, each of which reversed at least one finding.

Trap 1: cosine and probe metrics are duals of the training loss. Probe correlation and latent cosine rise mechanically when probe or slot losses are optimized—they measure whether information is *present*, not whether prediction *uses* it. Probe supervision lifts velocity decodability from $\rho=0.44$ to 0.80 and gives the most stable long-horizon *cosine* (0.882 vs. 0.843), while its *pixel* rollout is the worst in the comparison (19.93 vs. 20.64 dB at horizon 28). Appearance augmentation on Physion++ improves the aggregate long-horizon cosine while per-scene nMSE collapses up to $100\times$ (deform_clothhit: cosine 0.610 $\rightarrow$ 0.913, nMSE 0.772 $\rightarrow$ 3.69). Judging either result by direction-only or readout metrics inverts the conclusion. Our own early sweep was misled this way: judged on $K=4$ probe ρ , $\lambda=50$ “won”; re-judged on prediction loss, the ranking reversed.

Trap 2: nMSE denominators can degenerate. When the target sequence degenerates (a parabola ball exits the frame, target variance $\rightarrow 0$), per-horizon nMSE explodes— 3×10^4 – 2×10^6 at horizon ~ 28 across all six parabola baseline variants, while cosine sits at a healthy 0.55–0.95—and pollutes every aggregate that includes late horizons. This is why parabola is judged on its clean r/m -OOD partition throughout. Traps 1 and 2 point in opposite directions (cosine under-reports failure; degenerate nMSE over-reports it); neither metric family is safe alone, hence per-partition, per-horizon cross-validation.

Trap 3: zero-shot transfer is bounded by the architecture prior. On Physion OCP, a *randomly initialized* encoder scores mean AUC 0.607. No synthetic-training configuration exceeds it: free rollout approaches it (0.603), physics variants fall below it (0.551–0.582), appearance augmentation 0.579–0.597; training longer approaches the ceiling monotonically (0.554 at epoch 1 to 0.603 at epoch 20) without crossing. Per-scenario, only Support (+0.10) and Collide (+0.07) show signal above the random control. Zero-shot claims must be calibrated against the random-prior floor, or they measure the architecture, not the learning.

Trap 4: protocol confounds fabricate results in both directions. Three probe-protocol choices (random vs. pre-trained initialization, single-frame vs. stacked probes, probing through the projector) multiply into a false negative: the naive stack reads uniform velocity decodability at $R^2=0.166$, the corrected protocol 0.939. A silent initialization bug (only 24 of 216 loadable encoder keys actually loaded) invalidated an entire 45-configuration sweep before a loading guard caught it. And converged training loss proves nothing about rollout: our clean from-scratch baseline converges to the pretrained model’s prediction loss (0.006 vs. 0.005) while its rollout OOD error is $2.8\times$ worse.

Trap 5: loss re-weighting masquerades as physics. On the frozen-DINOv2 backbone, heavily re-weighting the

pinned position slot (weight 300; the 2 slot dimensions then carry 76% of the prediction loss) improves uniform both-OOD from 0.427 ± 0.039 to 0.286 ± 0.015 —apparently the one clear injection win on either backbone. Two content-free controls dissolve it: pinning the slot to a *shuffled*, physically meaningless target at the same weight reaches 0.330 ± 0.013 , and weakening the pin $10\times$ while keeping the weighting reaches 0.329 ± 0.084 —most of the gain, with no usable physics. All three variants degrade in-distribution error ($0.099 \rightarrow 0.115$ – 0.136), the signature of capacity restriction (the predictor is effectively reduced to a 2-dimensional task), and the same recipe *worsens* collision ($1.11\times$) and parabola ($1.14\times$). An apparent injection gain must be checked against a content-free control at matched loss weighting.

Auxiliary-loss dominance (measurement behind §4.3 of the main paper, Test 2). At probe weight $\lambda=10$, the weighted auxiliary-to-prediction loss ratios—used as a proxy for gradient magnitude; we do not measure gradient norms directly—are $15\times$ (collision), $44\times$ (uniform), and $125\times$ (parabola); the encoder’s effective (participation-ratio) dimensionality collapses by 39–90% relative to baseline (uniform 41 $\rightarrow$ 4). At $\lambda=1$ the ratios are benign, which is the setting used in §4.2 of the main paper.

Checklist. (i) Report per-partition *and* per-horizon; never aggregate across horizons without inspecting them. (ii) Judge with scale-aware prediction metrics (nMSE, decoded-pixel PSNR); use cosine and probe ρ as diagnostics only. (iii) Audit nMSE denominators for degenerate targets. (iv) Calibrate transfer against a random-init control, and validate probe protocols on a random encoder.

References

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